



CoTacs: A Haptic Toolkit to Explore Effective On-Body Haptic Feedback by Ideating, Designing, Evaluating and Refining Haptic Designs Using Group Collaboration

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ABSTRACT

Designing effective haptic feedback is challenging due to the subjective nature of touch and the fact that multiple people cannot easily share and evaluate touch experiences. In this work, we propose CoTacs, a collaborative haptic toolkit to address the challenges of haptic feedback design by enabling designers to explore haptic experiences together using group collaboration and allowing them to refine ideas quickly through early feedback during the design process. Two design sessions with five collaborators each show that CoTacs enables users to design, evaluate, refine and explore haptic feedback together. Participants leveraged group feedback to improve their designs and inspire creative ideas for the second collaboration round. Our work demonstrates how collaborative haptic toolkits can enable synergistic effects through interactions between the designers, benefiting haptic feedback design while still exposing several limitations. We discuss different opportunities for the design of future collaborative haptic toolkits and haptic group collaboration techniques.

KEYWORDS

Haptic feedback; tactile; touch; multimodal; design; toolkit; collaboration; extended; reality; group; tools; brainstorming

1. Background & introduction

Haptic feedback can provide more natural and intuitive feeling interactions (Kuchenbecker et al., 2006; Ramsamy et al., 2006; Ziat et al., 2014), reduce cognitive load (Leung et al., 2007; Meier et al., 2015; Oakley et al., 2000), improve task performance (Hoggan et al., 2008; Ma et al., 2015) and enhance the overall user experience in various applications (Bermejo & Hui, 2021; Elvitigala et al., 2022). In recent years, there has been a surge in the development of haptic feedback devices to bring these benefits to the emerging extended reality (XR) environments including augmented reality (AR), mixed reality (MR) and virtual reality (VR) environments. However, exploring effective haptic feedback for the new interaction environments is not a trivial task. In contrast to audio and vision, established design toolkits are yet missing and the unique characteristics of our sense of touch pose fundamental challenges for individual designers to come up with meaningful haptic designs that evoke the same experience in different people (Clark et al., 2017).

Three key characteristics of our sense of touch make haptic feedback design particularly challenging compared to audio and vision. (C1) Addressing our perception of touch is highly complex. Interfacing and stimulating the different types of receptors distributed across the body to evoke an intended sensation pose a significant challenge for technology as well as the haptic designer. The haptic designer needs

to explore how to use the supported capabilities of available haptic technology in order to develop a desired touch experience (Abraira & Ginty, 2013; Löfvenberg & Johansson, 1984; McGlone & Reilly, 2010). This is particularly the case for VR and MR environments where the technology often comprises a trade-off between expressive on-body haptic feedback and an unobtrusive wearable form factor. (C2) Unlike audio and vision, touch is discreet and intimate. It cannot be easily perceived simultaneously by multiple people through a shared medium. When e.g., producing music, multiple composers can listen to the same sound waves propagating through the air, allowing them to discuss and refine the musical experience together. The same does not naturally apply to haptic feedback, as direct contact to the skin is required. (C3) Touch is inherently subjective. The experience of touch can vary due different factors. Research has shown that individual differences such as age and gender-related differences play a role how well we perceive certain stimuli (Abdouni et al., 2018; Likens et al., 2020; Stevens, 1992; Woodward, 1993), and contextual factors as well as our past experiences influence our emotional response to them (Seifi & MacLean, 2013). This makes it hard for individual designers to come up with meaningful designs that are consistently perceived across different people without evaluating the experience with others first.

Collaborative haptic toolkits can help to address these challenges and support the design of on-body haptic

feedback at an early stage. They can provide the tools to explore and refine more expressive on-body haptic feedback for the new interaction environments and ease the process of finding a good match between supported output capabilities of the available technology and the target experience (C1). Collaborative haptic toolkits can enable designers to perceive haptic feedback simultaneously and allow them to share and discuss designs together like music composers do (C2). This way, they can furthermore enable early feedback about the haptic experience during the design process which allows to compensate for perceptual differences and produce more effective designs that are consistently perceived across different people (Demers et al., 2021; Schneider et al., 2016) (C3). Prior work has highlighted the potential of collaborative haptic toolkits by allowing designers to produce simple sketches and discuss haptic experiences together quickly (Schneider & Maclean, 2014; Wittchen et al., 2022). While they enabled shared experiences and discussions between multiple people, their approaches did not focus on how an evaluation process could feed into subsequent iterations of the haptic designs. We argue that collaborative haptic toolkits that specifically support designing, discussing, rating and subsequently refining haptic designs through a semi-structured process make shared haptic experiences enabled by collaborative toolkits more valuable and allow to explore haptic feedback designs more effectively.

In this work, we present CoTACS, a new collaborative haptic toolkit to support designers in exploring haptic feedback for XR environments more effectively by ideating, designing, evaluating, and refining on-body haptic experiences together. Our toolkit addresses the limitations and extends the work of Schneider and Maclean (2014) and Wittchen et al. (2022) by overcoming the “inability to tweak,” introducing a first group evaluation process, and allowing users to easily reuse and iterate on each other’s designs. We investigate the interactions and dynamics in two groups of 5 collaborating designers each, who use CoTACS to explore haptic designs together through a semi-structured process. Our findings reveal that our collaborative haptic toolkit could enable synergistic effects through the interactions between the designers during the collaborative design task. We show how CoTACS enables designers to reuse and iterate on each other’s designs and demonstrate how the rating of designs allows us to quickly identify better ones based on early group feedback. We discuss several limitations of CoTACS and share considerations for the development of future collaborative haptic toolkits and collaborative haptic group design processes based on our observations.

In summary, our contributions comprise:

- A new collaborative haptic design toolkit including a web-based GUI, a hardware controller board and a composite actuator to design and explore multimodal on-body haptic feedback.
- A first concept of a semi-structured collaborative approach to explore, design, evaluate and iterate on haptic feedback designs together.

- Insights from our evaluation with two groups of 5 haptic collaborators that help to better understand and improve haptic collaboration processes with collaborative haptic toolkits.

1.1. Outline

After discussing related work, we present the components of our proposed collaborative haptic toolkit and the intended supported collaborative process in detail. We explain how the individual features support collaborative haptic design and highlight the features that extend prior work beyond sharing to evaluating and refining haptic designs. Our toolkit features a new composite actuator to explore expressive on-body haptic feedback for the new XR environments. Therefore, we first verify the perceptibility of our newly developed composite actuator for rendering and exploring multimodal on-body haptic feedback in a first study. In a second study, we then investigate the usability of CoTACS and explore how our proposed toolkit helps address the challenges of designing haptic feedback by supporting, developing, evaluating, and refining haptic feedback designs together in two groups of five collaborators, each. Finally, we discuss our observations, look into current limitations, and share insights that can help inform future collaborative toolkits and haptic group design processes.

2. Related work

We identified four core concepts of our work: (1) new on-body haptic technology, (2) collaborative design and group, (3) haptic authoring and design tools, and (4) collaborative haptic toolkits. In the following, we discuss prior literature related to these four core concepts of our work (Figure 1).

2.1. On-body haptic feedback technology

Before the trend of mobile computing, haptic devices mainly comprised bulky stationary force feedback devices (Jarillo-Silva et al., 2009; Seifi et al., 2019). With the requirement of more compact feedback devices for mobile computing vibration motors have become the defacto standard to provide tactile feedback in mobile and wearable devices. They are highly available, compact, have low power consumption, and are easily perceivable on the skin. Linear Resonant Actuators (LRAs) are increasingly replacing Eccentric Rotating Mass (ERM) actuators as they provide more localized feedback and more control about the generated vibrations. Their implementation has been shown to enhance various applications such as to guide vision impaired people Hong et al. (2017), improving user performance for mobile interaction tasks Lee and Starner (2009), or conveying high bandwidth information through multi-modal feedback (Delbecq & de Ven, 1971) or spatial patterns Delbecq and de Ven (1971); Elvitigala et al. (2019); Hsieh et al. (2016); Luzhnica and Veas (2019).

However, vibrotactile feedback alone is perceived as artificial. It only suits to provide one of many haptic feedback

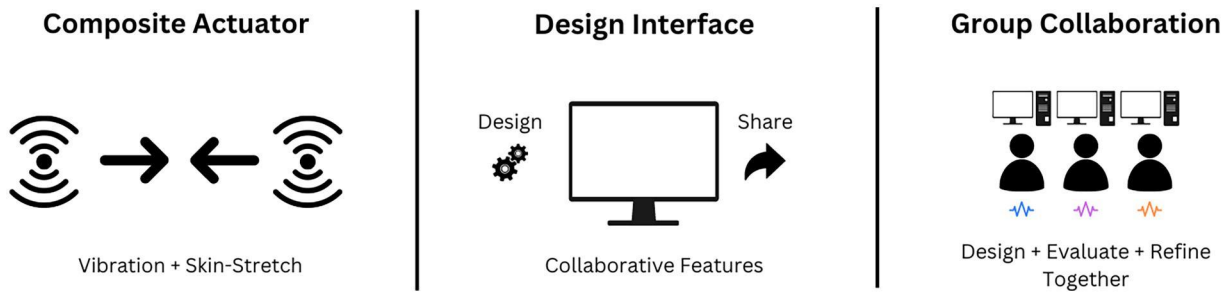


Figure 1. We developed CoTACS, a collaborative haptic toolkit comprising a programmable composite actuator and a design interface. The composite actuator can provide skin-stretch and vibration feedback and supports a broad design space to explore. The web-interface allows programming the actuator easily and supports collaborative features to support group collaboration between multiple designers. We show that collaborative haptic toolkits have the potential to support haptic designers to design, evaluate and refine haptic designs together to explore haptic feedback more effectively.

modalities that are required to generate more natural and intuitive feedback in MR and VR environments Huisman et al. (2013). To address the lack of expressiveness various haptic feedback devices have been proposed over the past two decades (Prattichizzo et al., 2013). While many of them can provide more expressive feedback, most exhibit bulky and rigid systems based on conventional actuators. This drove researchers to investigate more compliant wearable interfaces such as using electrodes (Lopes et al., 2017, 2018; Tanaka et al., 2023; Withana et al., 2018), dielectric elastomers actuators (DEAs) (Koo et al., 2008), twisted and coiled polymers (TCPs) (Chossat et al., 2019), and shape-memory alloys (SMAs) (Muthukumarana et al., 2019; Nakao et al., 2019).

Particularly SMA spring actuator have aroused great interest in the HCI community recently. In contrast to vibration motors and electro-tactile feedback devices, SMA actuators provide a smooth and silent contraction which can be utilized in order to physically deform the skin. Compared to DEAs and TCPs, SMA springs offer a compelling large actuator displacement of 50%. Shape-memory alloys have been incorporated into rings, straps, wrist bands and garments (Chernyshov et al., 2018; Compton et al., 2021; Gupta et al., 2017; Kim et al., 2023; Scheibe et al., 2007; Suhonen et al., 2012; Yarosh et al., 2017). Recent work has looked into producing more carefully designed stretch and pressure stimuli directly on the skin using the soft actuators. They demonstrate compelling thin on-skin patches that afford to be located at challenging body locations such as the ear, the chest or the foot ankle (Hamdan et al., 2019; Kim et al., 2021; Muthukumarana et al., 2020; Simons et al., 2020; Sun et al., 2020).

The skin-deformation stimulates mechanoreceptors that are sensitive to pressure (Merkel cells, SA-I) and stretch stimuli (Ruffini endings, SA-II) (Delmas et al., 2011). The resulting sensations can feel more natural than ones induced by vibration stimuli alone (Pacian corpuscle, RA-II) (Simons et al., 2020). Prior work showed that combining different modalities can allow to generate even stronger sensations than the vibrotactile on its own (Ion et al., 2015). Researchers have therefore highlighted that SMAs can be used as a promising complementary feedback technology to vibration.

In this work, we take up this idea and combine vibration motors with SMA springs to provide a compelling compact actuator unit to provide expressive on-body haptic feedback. Its adaptability to various body locations opens up a large design space that presents many opportunities for designers to explore haptic feedback applications within XR environments.

2.2. Haptic authoring tools & design platforms

Over the last two decades, a growing number of haptic authoring tools have been proposed for kinaesthetic force feedback and, more recently, focusing on vibrotactile feedback. One of the earliest proposed editors to design haptic experiences was *The Hapticon Editor* (Enriquez & MacLean, 2003). It allowed to record and replay small force feedback samples (hapticicons) analogue to visual and auditory icons. The user could also manually edit the waveform similar to conventional audio editing methods. Subsequent work proposed further interaction features like a tile palette and tile pane to compose haptic icons from basic haptic effects (Ryu & Choi, 2008; Swindells et al., 2006). With an increasing focus on vibration motors, researchers started proposing authoring tools supporting devices using multiple actuators and considering spatial actuator arrangement in the editor as a supporting programming metaphor (Kaul et al., 2019; Martínez et al., 2014; Panëels et al., 2013; Pezent et al., 2021). Several work focused on programming tools to augment audio and visual media content with haptic feedback (Eid et al., 2008; Swindells et al., 2014). While Swindells et al. particularly focused on augmenting dynamic 2D visual media content, Eid et al. proposed an authoring tool, *HAMLAT*, focusing on static 3D objects in virtual environments that can be rendered haptic-visually (Eid et al., 2008). With *HAMLAT* the authors furthermore aimed to simplify the design process for non-experts and allow them to focus on specifying defining haptic properties of an object instead of defining the raw actuation signal itself.

Also other work investigated more intuitive haptic design and programming methods. Hong et al. developed a tool relying on demonstration based programming (Hong et al., 2013; Ku et al., 2023). They argued that opposed to waveform editors which afford accurate design for small samples, demonstration based programming techniques are

advantageous when designing long dynamic haptic feedback. Lee et al. proposed the *VibScoreEditor*, which borrows familiar elements from music composition (the piano score and the guitar tablature) to design haptic feedback (Lee et al., 2009). *TECHTILE* comprises an intuitive educational toolkit to easily record and play tactile feedback (Minamizawa et al., 2012). Schneider et al. also investigated techniques that aim helping experts program vibrotactile grid displays more intuitively by direct manipulation with a visual interface (Schneider et al., 2015). With the help of the *Macaron* editor, the authors investigated how designers can leverage examples and simple operations to develop haptic feedback (Schneider & MacLean, 2016).

Particularly in the recent years, powerful prototyping toolkits have been developed to explore on-body haptic feedback applications on the skin using SMAs Messerschmidt et al. (2022); Muthukumarana et al. (2022), and other types of actuators Endow et al. (2021); Shtarbanov et al. (2023); Zhou et al. (2023). Our toolkit borrows the basic principle of Messerschmidt et al. (2022) to mount SMAs on plasters and extends it by integrating two vibration motors to build a compact actuator unit that provides more expressive multi-modal haptic feedback. To drive all components, we contribute a new driver board. The user interface of our proposed toolkit differs from prior work in several ways. Firstly, our programming interface allows easily programming and synchronizing SMA and vibration motor components to create multi modal haptic experiences. Secondly, it introduces collaborative features such as saving and loading shared designs, synchronous play and automatically hiding parameter controls to prevent visual bias during the group evaluation phase.

Most prior prototyping and authoring tools were developed to support the design and exploration of haptic feedback with a single user in mind and not targeting a collaborative design process itself. In the following section, we discuss tools and libraries focusing on features that enable sharing haptic feedback in different ways. They support aspects of collaborative haptic feedback design to different degrees.

2.3. Haptic libraries & collaborative design tools

Due to missing standards in haptics, sharing and collaborating on haptic designs is a challenging task and tools to enable collaboration between designers are only evolving slowly (Schneider et al., 2022). Over the past two decades there have been several approaches that either to support reusing existing ideas and designs between haptic designers through libraries or providing collaborative toolkits supporting an interactive collaborative design process directly.

Driven through early commercially available stationary haptic force feedback devices such as the *PHANTOM Omni Stylus* (Jarillo-Silva et al., 2009), prior work mostly focused on texture libraries that allow rendering textures of virtual surfaces or objects. Culbertson et al. developed a Penn Haptic Texture Toolkit (HaTT) with 100 haptic texture and friction models and provide the resources to render them

using available haptic consumer hardware (Culbertson et al., 2014). Next to consumer hardware there have been various hardware devices proposed by researchers. *Haptipedia* is a platform that aims to make the “fragmented” research landscape of haptic hardware designs and prototypes more accessible to researchers, enabling engineers and researchers to more easily to built up on top of existing ideas of hardware designs (Seifi et al., 2019). With vibration motors increasingly integrated in mobile devices and wearables such as smartphones and smartwatches, vibration pattern libraries and tools have been proposed more recently. With *VibViz* Seifi et al. provide a library to navigate vibrotactile patterns in a more accessible way based on functional and affective characteristics (Seifi et al., 2015).

Libraries can form a way to share and reuse haptic design ideas between designers and collaborate in a timely asynchronous manner. Focusing on the evaluation process during haptic designs more directly, Schneider et al. proposed crowdsourcing affective ratings for vibrotactile haptic designs (Schneider et al., 2016). While a general standard for the haptic feedback modality is not yet existing, the authors argue that the increasing amount of vibration motors integrated in mobile devices allows to crowdsource feedback from low-fidelity haptic proxies played at the end-consumer devices, that still allow to inform high-fidelity vibrotactile feedback designs. In contrast to this time and resource intensive evaluation approach for high-fidelity vibro-tactile designs, *mHIVE*, targets the collaborative design process for low-fidelity prototypes. It introduces a haptic instrument that allows improvising vibrotactile designs in real-time using a “haptic instrument” and exploring different designs (Schneider & Maclean, 2014). The authors tested and evaluated *mHIVE* in a qualitative way by interviewing designers improvising with the “haptic instrument” and allowing the interviewer experience what the designer designs at the same time. The authors highlight that this allowed the designer to better communicate with the interviewer, as the interviewer could feel the haptic feedback the designer was referring to during the interview. This approach focused mostly on a single vibro-tactile actuator worn by the designer and interviewer to share the improvised haptic feedback. Recently, Wittchen et al. introduced a more expressive collaborative toolkit, *TactJam*, to explore spatial vibration patterns through multiple body-worn vibrotactile actuators (Wittchen et al., 2022). *TactJam* features a software tool to quickly view and share designs, and a hardware kit to quickly develop low-fidelity spatio-temporal vibration patterns on the body using a hardware interface. They explore *TactJam* with 17 participants in two workshops and evaluate the toolkit in a qualitative way based on observations and interviews with the designers.

Our tool extends *mHIVE* and *TactJam* by proposing a new type of collaborative design toolkit. We address the “Inability to tweak” of *mHIVE* and give designers more control to improve their design. In *mHIVE*, users can quickly sketch ideas by sliding a finger along a 2D input field, which modulates frequency and amplitude in real-time. While this resembles the characteristics of an instrument for

improvisation, it requires re-sketching to refine an initial idea. In contrast, our system introduces GUI elements known from audio design software to control parameters via nodes along a specified time period. This allows users to draft initial ideas quickly but also refine them later based on feedback during the collaboration process. Initial ideas can be rapidly drafted in an intuitive visual way using parameter control elements borrowed from the audio design domain. Users can place a few nodes that define the parameter behaviour over time to resemble the initial design idea roughly. Later on, users can refine their initial idea by adjusting and adding more parameter control nodes. In addition to better control over the refinement, we introduce a semi-structured collaborative approach with an evaluation phase, where designers rate each other's designs as suggested by *TactJam*. In contrast to *TactJam* and *mHIVE*, we allow a group of 5 designers to work together simultaneously, enabling them to interact with each other and refine their designs after a group evaluation phase. We evaluate our toolkit in a quantitative way regarding two popular toolkit evaluation metrics and focus on the interactions between the designers over two iteration rounds in different collaborative settings. Our toolkit implements features to refine designs and allows the collaborating designers to iterate on each other's designs to create mid-to-high fidelity haptic feedback designs.

2.4. Group collaboration & techniques

Collaboration and group are often interwoven. Collaboration refers to the act of working together with others toward a common goal, whereas group is when a group of people come together to generate creative and innovative ideas. In order for group to be successful, collaboration is often necessary as it allows individuals to build upon each other's ideas, provide feedback, and work together to refine and develop the best ideas. Teamwork and effective communication are key elements in both collaboration and group. Many different group techniques have been proposed particularly in context of engineering design. To the most common group techniques belong Osborn's brainstorming (Osborn, 1953), the nominal group technique Delbecq and de Ven (1971), brainsketching (Van Der Lugt, 2002), C-Sketch/6-3-5 (Shah et al., 2001) and the gallery method (Linsey et al., 2005). Osborn's brainstorming is probably the most commonly known technique where groups of individuals brainstorm solutions to a given problem relatively freely together, with the exception of a set of restricting rules. There has been critique by subsequent work about the quality of the Orkan's original technique compared to nominal groups where individuals generate ideas independently and the ideas are discussed in a structured way (Bartunek & Murningham, 1984; Delbecq & de Ven, 1971). In the nominal group technique each person takes some time to silently generate ideas on their own. Then each person shares their ideas one at a time, and the group records them. Once all the ideas have been shared, the group discusses and evaluates each idea. The group then ranks the ideas in order of preference and selects the most appropriate one or ones.

Several studies indicated that nominal groups outperform real brainstorming groups mainly due to production blocking, i.e., individuals cannot focus on generating ideas while interacting with the other brainstorming group members (Diehl & Stroebe, 1987; Mullen et al., 1991). Other work showed that large group sizes can outperform nominal groups in computer aided brainstorming (Dennis & Valacich, 1993). Prior work highlights specifically how evaluation steps can help increase the quality by taking up constructive feedback but also making workers consider the task more serious Dow et al. (2012). Next to nominal groups several other more structured approaches to brainstorming have been proposed. In brainsketching and C-Sketch/6-3-5, group members first silently generate/sketch ideas. They then switch to someone's else idea and iterate on them. Ideas are here displayed in form of "Rotary Viewing", i.e., each participant views one idea after the other and not all at once (Linsey et al., 2005). Depending on the technique ideas are explained to the others or not. In the Gallery Method, participants first sketch their ideas and then discuss them together. A unique characteristic of the Gallery Method is that the ideas are displayed together in a "Gallery." Group members can walk around and study the gallery before the group starts another round of sketching.

While these collaboration techniques have been shown to be effective for engineering and visual design, it is yet unknown whether they can be adapted to the haptic design domain. In our work we chose to adapt a combination of the nominal group technique and gallery method for haptic group. This differentiates our work from prior work on collaborative haptic design toolkits. In our work participants discuss and rate one haptic design idea after the other according to haptic related criteria. They also have the ability to study haptic design ideas of others in detail and iterate on them based on previously received feedback.

This process is enabled through our developed toolkit, CoTACS, which we describe in detail in the following section.

3. CoTACS

CoTACS is a collaborative haptic toolkit that aims to support the design and exploration of on-body haptic feedback by enabling designers to design, share, evaluate, and refine haptic experiences together more effectively. It comprises three core elements: (1) A versatile composite actuator capable of rendering a diverse set of on-body haptic feedback for XR environments (Figure 2, left), (2) a programmable controller board that can drive the composite actuator (Figure 2, right), and (3) a web-based user interface that supports collaboratively designing and exploring haptic feedback together (Figure 3).

3.1. Composite actuator

To support expressive on-body haptic feedback for XR environments and provide an interesting design space to explore, we chose to design a composite actuator that provides vibrotactile as well as skin-stretch stimuli. It is motivated by

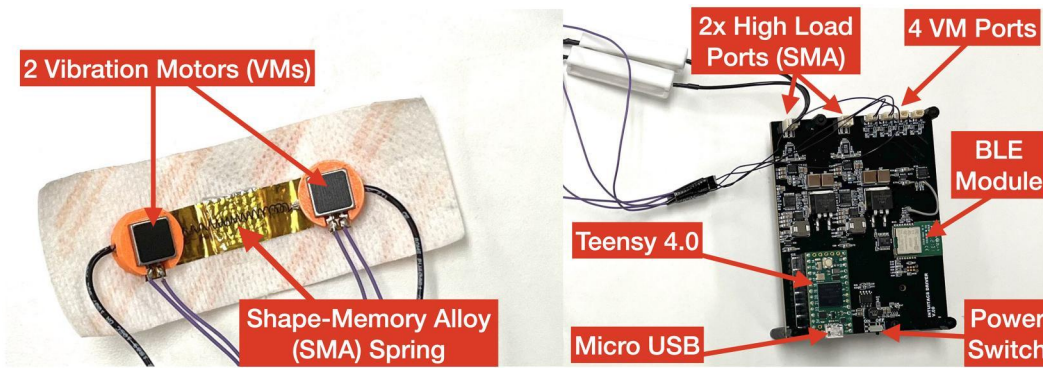


Figure 2. The composite actuator (left) features two vibration motors connected through a SMA spring to provide vibration and skin-stretch feedback. It can be applied anywhere on the body by sticking the plaster to the desired location. The controller board (right) can be programmed through our web-interface and drives the composite actuator accordingly.

recent research that suggests combining natural feeling on-body haptic feedback generated through soft SMA actuators together with abstract but more noticeable vibration feedback using vibration motors (Hamdan et al., 2019; Ion et al., 2015).

Our composite actuator (Figure 2, left) features two linear resonant actuators (LRAs) (VLV101040A) attached to the ends of a shape-memory alloy (SMA) spring (90-degree Flexinol Spring) to support spatial and temporal vibration patterns and provide lateral skin stretch via a compelling compact unit. Compared to other actuators, the VLV101040A supports a wider resonating frequency band around its central resonant frequency at 170 Hz with a high peak acceleration of up to 2.75g. This allows more precise control over a larger amplitude and frequency spectrum. The LRA's footprint comprises 1 cm², suitable for many on-skin applications, and has therefore been used by prior on-body haptic prototyping toolkits (Dementyev et al., 2021). Similar to (Hamdan et al., 2019; Messerschmidt et al., 2022)), we implement a 2.54mm Flexinol Spring Actuator with an elongated length of 40 mm which have shown to provide a compelling large displacement and strong contraction force, suitable to produce skin stretch stimuli at different body locations. We attach the three components to a plaster which can be easily stuck to the skin like (Messerschmidt et al., 2022). The SMA was shielded using a combination of Thermafect heat insulation tape and polyimide (Kapton) tape.

The design of the composite actuator offers haptic designers to either control individual components to e.g., produce vibration feedback or skin-stretch only, or blend them together by actuating different parts at the same time. By manipulating different parameters such as amplitude and frequency various actuation patterns can be explored. We validate the interplay of the different components of our composite actuator in a preliminary perceptibility study in Section 5, and illustrate how the designer can control different parameters in Section 3.4.

3.2. Controller board

We designed the controller board (Figure 2, right) to support driving our composite actuator and other devices.

It uses four haptic drivers (DRV2605L) to control up to four LRA or eccentric rotating mass (ERM) vibration motors and two H-bridges (MP6519) that support up to 5 amperes to control two high load actuators such as SMAs, electric motors or pumps. The board is powered by 3.7 V 700mAh lithium battery (bottom) and can be switched on and off using a dedicated power switch at the side. To control the logic of the board we use a Teensy 4.0, which furthermore offers a serial interface. Communication to the board is established by using the USB port of the Teensy and connecting it to a computer via a USB cable. The controller board features components to support more actuators and a bluetooth module (BT832F) for wireless communication in the future. When the controller board is connected to a computer it allows easily driving our composite actuator based on the specified design in our design interface – which we discuss in the next section.

3.3. Design interface

The design interface is hosted via a *Node.js* web server and accessible through a web browser. Designers open the web address, choose a nickname and connect to a CoTACS controller board plugged into the computer. After that, they can start designing and testing haptic feedback with other collaborators connected to the same web server using the design interface (tested with Google Chrome, Version 103.0.5060.114).

The design interface is divided into three sections (Figures 3 and S1–S3). In the top Section (S1) is the haptic design gallery where the user can load previously saved designs of themselves as well as the designs of any user connected to the web server. In the center Section (S2) the user can set and modify haptic design parameters to compose haptic feedback using the three components of the composite actuator. On the bottom (S3) the user can save and play the design locally or instantly share, play and test it together with other collaborators.

In the following, we first describe how haptic designers can design and test haptic feedback on their own. We then discuss how our design interface supports collaborative activities.



Figure 3. The web interface allows programming the composite actuator, sharing and testing different designs. It can be split into 3 sections (S1, S2, S3). S1 (the design gallery) allows the user to browse and load designs which are identified by the designer's (participant's) name and the design name. In section S2 the user can set the behaviour of the individual actuator components over a specified duration and number of repetitions (a). For each VM it can be specified how the amplitude and frequency develop over time (B). The resulting waveform is shown in the waveform view (C). Additional buttons allow quickly applying commonly used operations like mirroring or copy and pasting parameter settings (D). The user can furthermore specify the waveform and phase-shift. For the SMA the power can be specified over time. The load bar indicates if the SMA reaches its maximum actuation load for the set time period to prevent overloading it (E). In section S3, the user can either save the design and play the actuation for themselves (F) or send the design to everyone connected to the session and play it simultaneously for everyone (G).

3.4. Designing haptic feedback

In contrast to Schneider and Maclean (2014); Wittchen et al. (2022), our design interface is designed to afford high control over different actuation parameters, enabling collaborators to iterate and refine their own or others' designs. Similar to prior haptic design toolkits, it borrows

design elements from the audio domain that have been proven to be suitable and easy to use for the haptic domain too.

The design Section (S2) allows the user to control key parameters for Vibration Motor 1 (VM1), Vibration Motor 2 (VM2), and the SMA Spring of our composite actuator. The parameters are displayed vertically (above each other)

to allow easily aligning the temporal behavior across the different actuator components.

For each component, the user can specify how the amplitude (0 – 100% duty cycle, $\sim 0g - 2.75g$ acceleration), frequency (0 – 170Hz) (VM1 and VM2), or power ($\sim 0 - 1.4A$, $\sim 0 - 320.95g$) (SMA) develop over time to alter the actuators behaviour. Similar to previously proposed haptic editors (Enriquez & MacLean, 2003; Schneider & MacLean, 2016) we plot a graph for each key parameter (amplitude, frequency, power), displaying how the parameter develops over a certain period of time (Figure 3 (B)).

The time period can be set by using the *Duration Input Field* (Figure 3 (B)) (100 – 10,000 ms). Furthermore, the user can specify how often the defined actuation should be repeated (1–20 repetitions).

The user can manipulate the parameter graph by clicking and dragging nodes (red squares). The signal is interpolated linearly between two neighbouring nodes. Using a left click a new node can be placed with a certain value for the parameter (y-axis) at a given time (x-axis) in the graph. Using a right click a node can be removed. Existing nodes and lines can be dragged in the graph to a different position. The parameter graphs can furthermore be manipulated using the buttons on the right (Figure 3 (D)) which e.g., allow mirroring the signal horizontally or vertically. A user can furthermore copy the graph of one parameter and paste it into a different parameter graph. The user can view the resulting waveform in a dedicated *Waveform* output graph below the parameters (Figure 3 (C)). To enable a more easy alignment between different actuators, individual parameter controls can be collapsed if desired. For vibration motors, the user can choose between five standard waveform types (sine, square, triangle, ramp, inverse ramp) and set the phase shift (0–360 degree). The latter shifts the waveform signal horizontally in time.

Since the SMA can not be controlled as precisely as the vibration motors and its output is hard to predict, there is no output signal displayed. Instead a rough indicator of the load is indicated with help of a progress bar. When the progress bar is full, the maximum load is reached. If the set power exceeds the allowed load, the progress bar is coloured red and the user is not able to play the design using the actuator. To further prevent the user from overloading SMA actuator accidentally, time and power constraints are implemented. The user can e.g., only actuate the SMA for one second at maximum power, or two seconds at half the power similar to Messerschmidt et al. (2022).

To play and test a design with the composite actuator, the user simply presses the play button on the bottom left (Figure 3 (F)). The play button will be disabled for the period the actuation is played, including an additional cool-down time of 10 seconds if the SMA is used.

3.4.1. Supporting collaborative haptic design

CoTACS's design interface implements several features to support collaboratively designing, evaluating and refining haptic feedback.

3.4.1.1. Sharing and feeling haptic designs together. The design interface allows to instantly exchange a currently configured design with all collaborating designers connected to the server upon pressing the “Send Design to Everyone” button. Collaborators can then either edit or play the design and share it again. To allow collaborators feel a haptic feedback design simultaneously and enable discussions about the shared touch experience, a designer can press the “Play For All” button. The currently active design is then played for all participants. If one designer wants to share and play their haptic design for everyone, they would hence, first send the design to everyone, and then play it for all. The collaborators would then feel the haptic feedback design and could immediately start making modifications if they wanted to. During pilot tests, we noticed that designers are easily biased through viewing the parameters and waveforms before experiencing the haptic feedback. Therefore, when a design is shared to everyone, all parameters and waveforms are hidden automatically. A designer can unlock and show the parameters through a simple button click at the bottom of the interface.

3.4.1.2. Evaluating & rating designs. After sharing and feeling a design, designers can discuss the haptic experience together. Each of the designers can then quickly rate a design by clicking on the “Rate” button. Upon clicking the button participants are presented with a form that allows them to anonymously rate the design based on two evaluation criteria as proposed by prior work Zhou and Sareen (2020). Participants first rate the design based on how well it fits the intended application scenario (*Goodness of Fit*) on a 5-point scale (1 not fitting at all – 5 perfect fit), and then rate the overall feeling of the design independent of how well it fits to the application scenario (*Goodness of Feel*) on a 5-point scale (1 not feeling good at all – 5 best feeling).

3.4.1.3. Reusing and refining designs. To enable collaborating designers to easily reuse and iterate on each others haptic designs we introduced an interface section which we call the *haptic design gallery*. While multiple haptic feedback designs can not be perceived at the same time similar to a visual gallery, we borrow the metaphor of a gallery where designers can go around, inspect each others ideas, and reuse and reiterate on them, similar to the gallery group method (Linsey et al., 2005). In our interface (Section 1), designers can easily inspect and modify designs by first selecting the corresponding designer name from the *Participant* drop-down box, then choosing the name of the saved design in the *Design* drop-down box, and finally pressing the *Load* button. Any saved modification will be saved under the modifiers name. Overwriting a design is prevented.

4. Haptic group collaboration with CoTACS

With CoTACS we propose a first semi-structured haptic group collaboration process complementing the toolkit. It involves ideating, designing, evaluating and refining haptic

designs together (Figure 4). The process is defined by two phases that are repeated for a specified number of times similarly to common group techniques.

During the first phase, the collaborating designers ideate and design a haptic experience on their own within a limited period of time of three minutes. They first come up with an application scenario and develop a preliminary haptic feedback design using the toolkit. After three minutes the collaborators present, discuss and rate their designs one by one. During the presentation, the designer first explains where to place the composite actuator, describes the application scenario, and then shares and plays the haptic feedback design. After that the collaborators discuss what they felt and comment on the haptic experience. Finally each collaborator rates the design regarding the *Goodness of Fit* and *Goodness of Feel* similar to Zhu et al. (2020). Whereby the *Goodness of Fit* measures how well the haptic experience fits the intended application scenario on a 5-point scale (1 not fitting at all – 5 perfect fit), and *Goodness of Feel* measures how good the haptic experience feels independent how well it fits to the application scenario on a 5-point scale (1 not feeling good at all – 5 best feeling). After this process the haptic designers can choose to either discard their design and start a new one, or refine their design based on the feedback. They can also use the haptic gallery to access and iterate on designs of other designers. This process can be repeated for a specified number of times. After the exploration process, a list of evaluated designs can be derived with a scoring that helps to identify the best designs to work with.

5. Study 1: Composite actuator perceptibility

Before facilitating a collaborative haptic design session with CoTACS, we conducted an initial preliminary study to validate the perceptibility of our new proposed composite actuator and ensure that haptic designers would be able to explore different haptic experiences. We wanted to see if the wearer would be able to feel differences when actuating different elements of the composite actuator individually, or in combination.

5.1. Participants

We invited 10 participants (7 male, 3 female) aged between 20 and 35 ($M = 27.4$, $SD = 4.09$). Five had prior experience with haptic feedback other than vibration.

5.2. Design & method

The participants sat down at a table, operating a computer with the dominant hand and the actuator mounted along their lower arm of their non-dominant hand. Users wore noise cancelling headphones playing white noise. The device was visually shielded by a cardboard box.

Participants went through 102 trials including 3 test trials. For each trial the participant experience one vibration pattern consisting of a combination of SMA Power Values (3 levels, no squeeze, medium squeeze, strong squeeze), Vibration Power Values (11 levels, 0–100%). Each combination was repeated three times. The full within-subjects design comprised 3 SMA Power Values $\times$ 11 Vibration Power Values $\times$ 3 Repetitions $\times$ 10 Participants = 990 trials excluding test trials. All trials were randomized. For the three test trials at the beginning, first vibration was played at maximum power alone, for the second trial the squeeze was played at maximum power alone followed by the third trial where both vibration and squeeze were played at maximum power together.

After each trial participants were asked to answer the noticeability of the vibration on a 7-point scale and the noticeability of the squeeze through the SMA on a 7-point scale (1 not noticeable at all, 7 highly noticeable).

5.3. Results & discussion

We plotted the dependent variables over the vibration power grouped by the squeeze intensity level in Figure 5. We could make the following observations:

1. We could see that noticeability of the vibration linearly increased with increasing vibration power (Pearson Correlation $r = 0.99$, $p < 0.000$, two-tailed). (5, left) The squeeze level did not appear to significantly effect vibration noticeability (Kruskal-Wallis $H = 1.308$, $p = 0.52$).
2. Participants could clearly feel the 3 different squeeze levels. A Kruskal-Wallis test revealed significant effect between Squeeze Level and Squeeze Noticeability ($H = 261.599$, $p < 0.001$). A Dunn's test confirmed a significant difference between the means of all squeeze conditions with $strong_squeeze > medium_squeeze > no_squeeze$ and $p < 0.001$ for all pairwise comparisons. The average squeeze noticeability ratings were $M = 1.045$ ($SD = 0.323$) for No Squeeze, $M = 2.179$ ($SD = 1.49$) for Medium Squeeze, and $M = 5.376$



Figure 4. Proposed haptic group collaboration process with CoTACS. It consists of two phases. In the first phase (3 min) collaborators ideate an application scenario and design a suitable haptic experience. In the second phase (10 min) designers share, discuss and rate their designs together as a group. The process is repeated for a specified number of times.

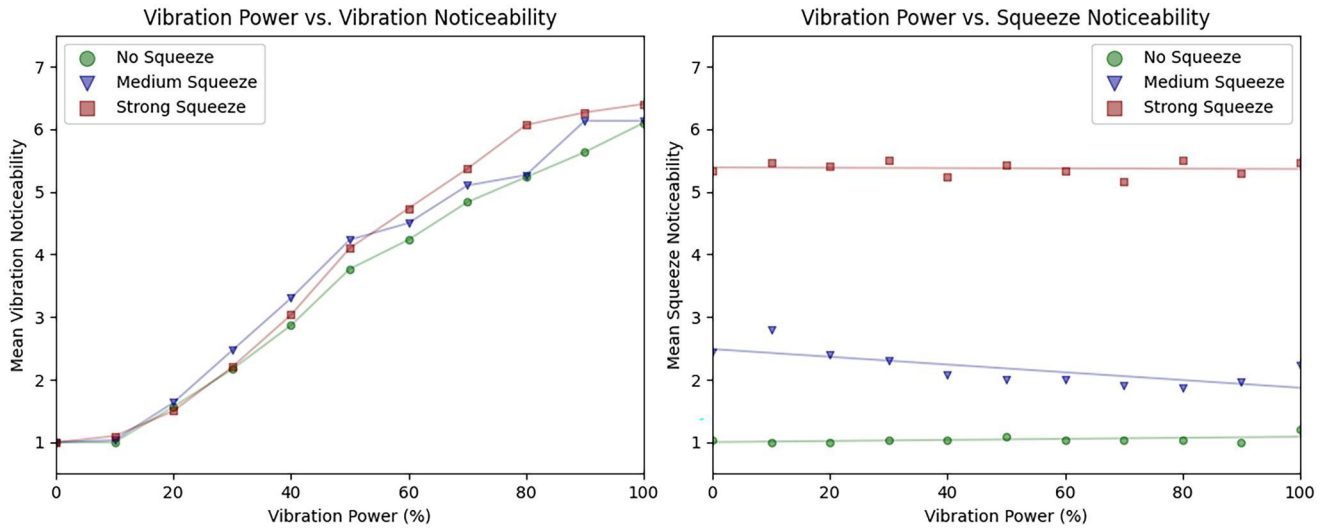


Figure 5. Two plots showing different parameters over Vibration Power (in percentage) for three conditions no squeeze, medium squeeze, strong squeeze: (left) mean subjective rating of vibration noticeability (1 not noticeable at all – 7 highly noticeable) plotted over vibration power. (right) Mean subjective rating of squeeze noticeability (1 not noticeable at all – 7 highly noticeable) plotted over vibration power.



Figure 6. Participants testing and feeling a design together that should provide a head collision warning using vibration motors mounted at the forehead.

($SD = 1.352$) for Strong Squeeze. Pearson correlation test did not show any statistical significant relation between vibration power and perceived squeeze noticeability (Figure 6).

Overall, the components of our composite actuator were easily perceived and participants could notice different levels of vibration and squeeze feedback. Based on the results we were confident that our actuator works and users would be able to manipulate the output of our actuator to explore different haptic experiences.

6. Study 2: Collaborative haptic design with CoTacs

In our second study, we wanted to explore the ability of CoTacs to support the haptic feedback design process. On the one hand, we wanted to investigate the general usability of our toolkit and verify that different designers can work with the tool and produce shared haptic experiences. On the other hand, we wanted to understand if our toolkit can help address the challenges of haptic feedback design by providing a more effective exploration of different designs through

a collaborative process. We sought to answer the following questions:

- Q1: Can our haptic design toolkit be used effectively by individuals with varying levels of technical expertise?
- Q2: Does our toolkit enable shared experiences between designers? (C2)
- Q3: Does our toolkit support a creative exploration of expressive on-body haptic feedback? (C1)
- Q4: Can we incorporate early group feedback during the design process to compensate for subjective differences and allow to refine the haptic experience? (C3)

6.1. Participants

We conducted two distinct study sessions with 5 participants in each session. In total, there were 10 participants, aged $M = 26.55$ years ($SD = 3.4$), 3 female, and 7 male. The demographics of the two sessions were slightly different. *Session 1* comprised 5 male participants, all researchers working in the area of VR. Two had prior experience with designing haptic feedback using vibration motors. One had

prior experience with SMAs, too. *Session 2* consisted of 3 female and 2 male participants. 2 of them did not have experience with VR. And 1 participant had prior experience with designing haptic feedback using vibration motors.

6.2. Design & method

Every participant was equipped with a laptop, a headset, one composite actuator and skin-adhesive spray. The composite actuator was connected to the laptop running our developed web interface in a Chrome web browser. Participants were spatially separated and communicated via the video conferencing platform Zoom. The study was divided into three phases. An introductory phase, a practice phase and a collaborative design phase.

- **Introduction Phase:** In the Introductory Phase, the participants were briefly introduced to the composite actuator. They were explained how the SMA spring works, how to use the two vibration motors, and the correct way to attach the actuator to the body. Afterwards, the researcher explained how the actuator can be programmed using the software interface by going through the individual elements one by one. The participants were furthermore shown how to play their design, how to share it and how to load designs of other participants.
- **Practice Phase:** After the introduction, the participants were asked to perform three practice tasks to get in touch with the different elements of the toolkit. For each practice task, the instructor displayed a prepared design to the participants and asked them to try to replicate and test it using the toolkit. The first practice design only made use of the two vibration motors. While the amplitude of VM1 increased slowly, the amplitude of VM2 decreased slowly.

In the second practice task the actuator played two short vibration pulses, first with vibration motor VM1, then with vibration Motor VM2.

The last practice task made only use of the SMA component and actuated it with two pulses. Each task ended when the participants signalled to be done by raising their “virtual hand.” The practice tasks also helped the instructor to ensure hardware and software were working properly. After completion participants were invited to play around with the software and hardware.

- **Collaborative Design Phase:** The collaborative design phase consisted of two rounds of the design process as described in [Section 4](#). In short the collaboration rounds were as follows:

Round 1: In the first collaboration round all participants were asked to design a haptic sensation for a VR scenario of their choice using the toolkit. They were given the freedom to use any combination of the composite actuator, utilizing only VM1, VM2, or the SMA alone, or using them together. After 3 minutes, everyone was asked to present their design. Participants could share comments on what they felt and rate the design regarding the *Goodness of Fit* and *Goodness of Feel* similar to Zhu

et al. (2020).

Round 2: The second collaboration round was designed exactly like Round 1, but participants were invited to either create a new design themselves or iterate on an existing design of any participant from the preceding round.

After the Collaborative Design Phase, participants rated the system based on the System Usability Scale (SUS) and the Creativity Support Index (CSI). Additionally, participants answered open-ended questions to gain further insights about the design process, its challenges, and potential improvements. Questions included inquiries about the challenges faced during collaboration, elements that aided in the process, and aspects that could further improve their work efficiency.: “What were the challenges you faced when working with each other?,” “What things helped you the most to work with each other?,” and “What could have helped you to work more effectively with each other?” We used participants’ responses to these questions, along with the SUS survey, to gain first insights about the toolkit’s usability. The CSI survey provided a quantitative metric to measure the toolkit’s supported creativity and exploration. Additionally, we examined the developed designs about the variety of used elements and parameters to analyze the design diversity. The development of the designs and the interactions between the designers were tracked via the design ratings, audio and video recordings. This way we investigated whether and how our toolkit enables designers to work together to explore, discuss, and refine haptic designs.

6.3. Results & discussion

One complete study session took about 80 minutes. Each of the three practice tasks took $M = 3.83$ ($SD = 0.74$) minutes to finish. In each session, the participants spent around 25 minutes creating and discussing haptic feedback during the Collaboration Task. Each discussion round (after Round 1 and Round 2) took $M = 14.15$ ($SD = 5.87$) minutes, depicting about 2.83 minutes of discussion time per design, including the design’s rating. At the end of the study, 20 different haptic feedback designs were produced ([Figure 8](#)) that had been discussed collectively (10 designs in each session). During the first session, we encountered one data recording issue that is that the design from P1 in the second round was not saved correctly.

6.3.1. Usability

After the practice tasks, most participants showed to operate the tool confidently. The system was overall rated to be easy to use ($M = 4.4$, $SD = 0.7$) (see [Figure 7](#)). Our system scored 73.0 out of 100 on the System Usability Scale (SUS), which is classified as acceptable and rated as “Good” based on (Bangor et al., 2008). Almost all participants (9 of 10) stated that the most helpful feature was instantly sharing and testing designs of others “I liked being able to send to everyone

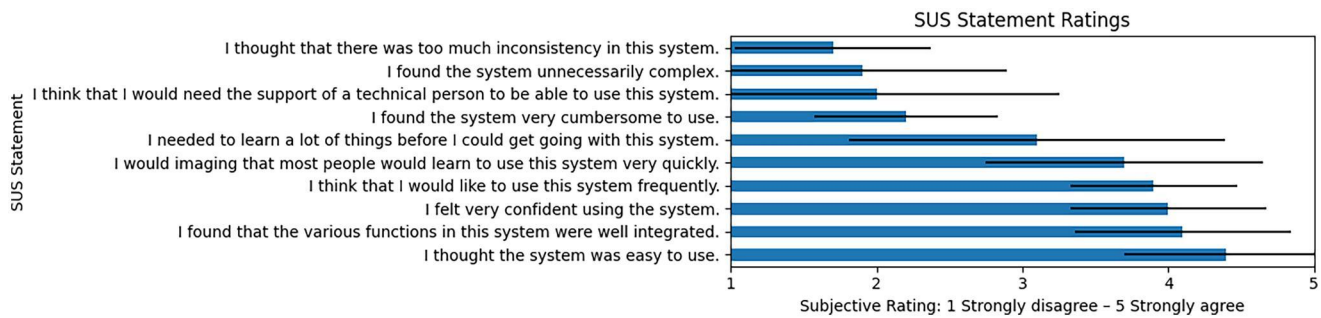


Figure 7. Mean subjective ratings of negative and positive SUS statements on a 5-point Likert scale (1 strongly disagree – 5 strongly agree) with standard deviation. The overall SUS score depicts $M = 73.0$ out of 100. Lower values for the top five questions higher values for the bottom five questions are interpreted as good..

PID	Design	Application	VM1	VM2	SMA	Duration	Rep.	Rating
1	1	Flight Simulator Landing				1000	5	3.00
1	2	Flight Simulator Landing				1000	5	4.63
2	1	Nudging				1000	1	3.75
2	2	Nudging				2000	1	4.00
3	1	On-skin Stroking Gesture				1500	1	3.50
3	2	On-skin Stroking Gesture				1000	4	3.63
4	1	Getting Hit by a Gunshot				2000	1	3.25
4	2	Obstacle Warning				503	5	3.88
5	1	Warning Notification				1000	5	3.50
5	2	Warning Notification				1000	5	3.88
6	1	Animal Crawling on Skin				1000	1	3.00
6	2	Firing a Gun				1000	5	4.50
7	1	Cold Breeze				2000	2	3.13
7	2	Massaging 2				500	4	3.25
8	1	Bee Landing on Arm				500	5	3.83
8	2	Gun Shot Impact				1000	1	3.50
9	1	Massaging				2000	3	4.33
9	2	Running Pace Feedback				500	5	3.67
10	1	Directional Navigation Cue				500	5	3.67
10	2	Touching Water				1000	1	4.13

Figure 8. Designs of all 10 participants. Displayed are the application, all parameters and overall rating of each design. Rating values are the average values of the mean Goodness of Fit and Goodness of Feel ratings (1 lowest, 5 highest). *Note:* Design 2 of PID 1 was not saved correctly.

to test and actually feel what others are doing and not just hear what the purpose was.” (P5). P2 furthermore highlighted that “Loading pre-existing designs by other team members was very useful and editing other’s design was a plus.”

For some of the participants who were less familiar with audio processing elements, such as waveforms and their manipulation, the learning curve was initially steeper. “I think its very simple for people who have knowledge about

waveforms, but would be very confusing if you had not done that before.” (P1). This is also reflected in the individual SUS ratings. “I needed to learn a lot before I could get going” was rated relatively high compared to the other negative SUS statements with $M = 3.1$ ($SD = 1.29$), and also having the highest dispersion.

Overall participants appeared to be able to work well with the implemented features. However, we noticed that it was sometimes challenging for the designers to communicate the

body location where the actuator was meant to be placed for the application scenario. A way to specify the intended body location of the actuator for a corresponding application could have helped convey it more easily during our remote collaboration setting.

In addition, we observed that minor usability flaws could easily disturb the flow of the discussions, possibly impacting output quality. In our case, an unexpected behaviour of our number input field which was meant to restrict the duration and repetition values within reasonable range confused some participants. While on an individual basis this would have been a minor issue, we recognized that when participants faced this issue the first time during the collaboration task, it could easily disturb the overall flow affecting the communication and engagement of the collaborators.

Finally, some participants wished that they had a second set of the composite actuator for their application scenario.

Q1: Can our haptic design toolkit be used effectively by individuals with varying levels of technical expertise? – In our study, users of different expertise could operate our system well within a short amount of time even though there was a steep learning curve for some of the participants at the beginning. The usability of our system was rated to be good. We gained insights how to improve the usability of our toolkit, particularly with respect to making the flow in a collaborative setting more friction less.

Q2: Does our toolkit enable shared experiences between designers? (C1) – Our toolkit could enable shared haptic experiences between designers and the participants highlighted the importance of the sharing feature for validating haptic experiences that served as basis for the group discussions.

6.3.2. Exploration & supported creativity

During the design tasks, participants developed 13 distinct applications, ranging from abstract tactile cues such as Obstacle Warnings (P4) to VR human-to-human gestures that emulate On-skin Stroking (P3) and even ambient touch sensations like feeling a Cold Breeze (P7) or the sensation of Touching Water (P10). We display the developed designs in Figure 8. Participants utilized the full range of parameters available to them in order to create the desired haptic experiences. They manipulated amplitude, frequency, duration and repetitions; adjusted the waveform, synchronized patterns and consistently used all three components of the composite actuator (vibration motor 1, vibration motor 2, and the shape-memory alloy spring) to create abrupt sharp or continuous smooth sensations corresponding to the application scenario. Most designs focused on synchronized temporal patterns, such as the Flight Simulator Landing, where both vibration motors were actuated exactly the same way. However, multiple participants also developed spatio-temporal patterns such as the Animal Crawling on Skin, or Directional Navigation Cues, where the spatial arrangement of the vibration motors was used to e.g., indicate a direction. Participants focused predominantly on the vibration motors. In many cases, the SMA was used to support the vibration motors, particularly with the aim to make the sensations feel

more natural. For very abstract haptic feedback such as the Warnings, participants completely omitted the SMA. We believe an additional factor why participants primarily focused on the vibration motors and only used SMAs to provide a complementary feedback modality, was that the composite actuator and programming interface allowed integrating vibration motors in a more dynamic way compared to shape-memory alloys.

Participants showed to be mostly satisfied with their outcomes. Results Worth Effort was rated high with $M = 15.60$ ($SD = 2.33$) out of 20 (see Table 1). In addition, other participants rated the designs on average medium to high, with an average Goodness of Fit rating of $M = 3.72$ ($SD = 0.90$) and an average Goodness of Feel rating of $M = 3.71$ ($M = 0.85$) out of 5.

From the five CSI criteria, Exploration was prioritized highest with 3.70, followed by Collaboration and Expressiveness (2.50 each). The lowest priority was given to Immersion (2.00). Similarly Exploration scored highest with the overall average weighted factor score with 61.60, followed by Collaboration (39.40) and Expressiveness (38.80). Immersion also scored lowest for the average weighted score with 31.90.

Q3: Does our toolkit support a creative exploration of expressive on-body haptic feedback? (C3) – Overall, our system showed to support a high degree of exploration and creativity when collaboratively designing haptic feedback. The participants showed to be able to develop a diverse set of haptic feedback designs. Based on the participants ratings our toolkit received a CSI score of 80.2 out of 100, which corresponds to a “B” grade.

6.3.3. Collaboration dynamics

During the collaboration task, participants chose to develop their haptic designs in different ways. For the second round, 4 participants decided to create a completely new design, 4 decided to reiterate on their own design, and 2 decided to adapt the design of others. In the following, we describe the unique ways participants developed their designs, which rooted in the interactions between the participants:

- Iterating: Improving on one’s own haptic design based on group feedback: P3 wanted to create a swiping gesture over the skin in a collaborative VR scenario. The device was placed on the wrist perpendicular to the arm, like a wrist band. During the discussions, P3 noticed the gesture was not well perceived by all participants. “[The design] does not feel like it was a swipe gesture to me.” (P5). Most participants could not identify a direction at first. P3 therefore decided to reiterate the design for the second round. He refined the amplitude and the SMA power, added frequency modulation, and let the design repeat four times to make the swiping more obvious. “[The swipe gesture] feels much more obvious.” (P5)
- Iterating: Improving someone else’s haptic design: P4 decided to iterate on the design of P5, which was intended to give people a tactile warning notification. The actuator was placed at the forehead. P4 wanted to

increase noticeability: “I thought if it’s a warning you must feel it more heavily. I made it switch more times so you better notice it.” (P4). He increased the frequency, added two additional impulses using the amplitude, and decreased the duration. “The increased amount of switching sides was good.” (P5)

- **Piggybacking:** Reusing someone else’s idea to achieve a haptic design goal: P9 designed an application for massaging using a constant vibration frequency and amplitude in the first design round. P9 suggested multiple body locations such as the head temple, foot, or neck for this application. In the second design round, P7 chose to focus on the same application, but wanted to modulate vibration frequency and amplitude. Instead of iterating on P9’s design, P7 chose to modify the design created by P8, which originally focused on a completely different application. P7 used P8’s design, increased the frequency, kept the SMA power constant, and changed the repetition number to 4.
- **Leapfrogging:** Remixing different ideas to create new haptic feedback designs: P8 decided to create a completely new design. While her initial design aimed to simulate a “bee landing on the arm,” she wanted to simulate “getting hit by a gunshot” for the second scenario. She was not satisfied with her quick draft, but other participants shared ideas on how else the sensation could be used linking it back to the first discussion round: P8: “I quickly had to redo it. I did want to make it sharp but it didn’t work out that well.” P6: “yes but it could actually be used as an extension to your previous design, as a sting of a bee.”

Figure 9 summarizes all observed pathways participants chose to come up with or propose new design ideas. These instances indicate that our toolkit could enable emergent effects like Piggybacking and Leapfrogging when designing haptic feedback, which can lead to more effective solutions. When analyzing the ratings, the designs from the first collaboration round were rated 3.52 ($SD=0.94$) for Goodness of Fit and 3.45 ($SD=0.79$) for Goodness of Feel. The designs from the second collaboration round were rated 3.88 ($SD=0.87$) for Goodness of Fit and 3.91 ($SD=0.86$) for Goodness of Feel. While the design ratings of the second round in our study were higher which would be an indicator that the designs of the second iteration were improved or better, we did not find a statistically significant effect. It is furthermore possible that the higher scores were due to order effects, which our study design did not account for. Hence, we could not infer any quantitative effect of the interactions between the participants, which were enabled through our toolkit, on an improved quality of the designs in the second round.

Q4: Can we incorporate early group feedback during the design process to compensate for subjective differences and allow to refine the experience? (C2)—We could observe instances (e.g., P3s design) where the group feedback informed changes to the first design and lead to an improvement in the second round. This was confirmed

through feedback from participants in the second discussion round. However, our results do not show a statistical significant effect when looking at all designs to generalize this finding. What we could observe is that several interactions between the participants inspired new creative ideas for the subsequent round.

6.3.4. Addressing the challenges of haptic feedback design

Overall our study showed that CoTACS can enable groups of designers to design, evaluate and refine haptic feedback together. The participants were able to use the toolkit to create a diverse set of haptic feedback designs for XR environments and explore different ideas. Despite several improvable elements, our toolkit yielded a good SUS score and the high rating of Exploration as well as Results Worth Effort indicate that CoTACS was able to provide the means to explore expressive haptic feedback for target the envisioned application (C1). In both sessions, the participants were able to share and feel haptic experiences together with the help of the toolkit, which enabled valuable discussions between five designers about the designed haptic experiences during the design process. We could observe how these discussions could lead to individual improved designs, which showcase the potential of collaborative haptic design toolkits to design haptic feedback more effectively (C2 + C3). Nevertheless, our toolkit still exhibits several limitations and leaves open several avenues to explore for future work. Especially in terms of quantifying the added value through CoTACS and similar toolkits. We discuss these limitations in the following section in greater detail.

7. Limitations & future work

In the following, we discuss the key shortcomings of our proposed toolkit and semi-structured collaboration process. We provide several suggestions how those shortcomings could be addressed in the future. We ground these in our own observations and qualitative feedback from the participants who shed light on positive as well as negative aspects of our approach.

7.1. Toolkit

7.1.1. Common ground to better understand and design for the application context

In our approach, participants thought about an application scenario and were asked to verbally describe it to the collaborators. However, quickly understanding the application context for a design idea was hard and often consumed larger part of the initial discussions: “[.] asking the scenario in detail and imagining it was a bit time consuming [.]” (P2). Means to establish and convey the application scenario more easily were yet missing, but could allow collaborators to discuss and evaluate the design based on a shared understanding of the context more easily. Haptic feedback is often designed complementary to other feedback modalities in the application context such as audio and vision. Being able to

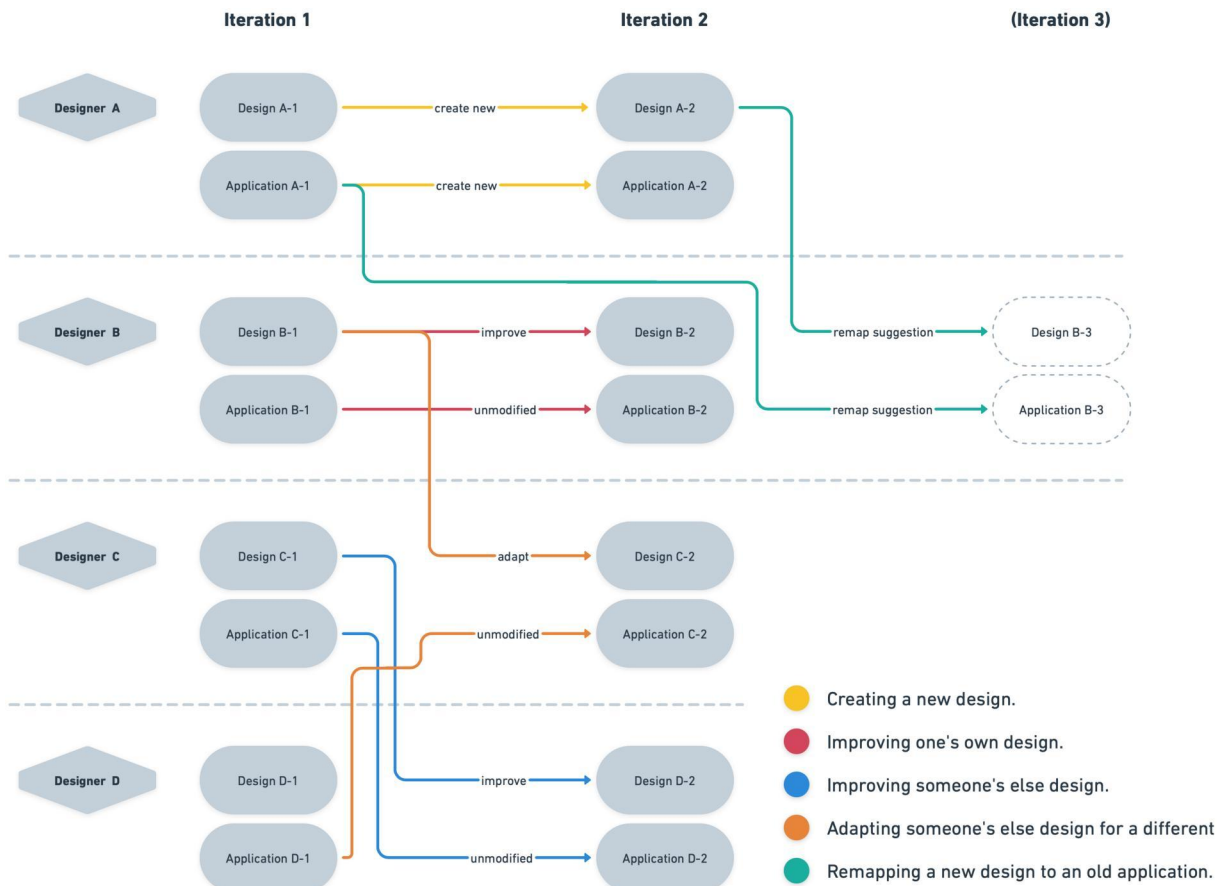


Figure 9. Unique pathways the participants chose to develop their haptic design ideas. Each unique pathway is represented by its own colour. Iteration 3 is based on a suggestion by a participant in the discussion round after iteration 2 and only hypothetical.

link and play haptic feedback in parallel to other content could allow testing and tailoring it directly within the application context (Eid et al., 2008; Swindells et al., 2014). Being able to share haptic designs linked to the application context would automatically provide common ground to understand the application context. We see features supporting (1) to play other media content, (2) to link the haptic feedback design to a specific media timestamp or event, and (3) to share the haptic feedback design linked to the media content, as future improvements that could enable to better understand and design for the application context using a collaborative haptic toolkit.

7.1.2. Specification of body location

Explaining other collaborators where and how the composite actuator was attached for the application context turned out to be cumbersome during our study. Our toolkit did not allow to specify the location. A feature that helps participants specify and share the actuators position on the body, similarly to Wittchen et al. (2022), would simplify the exchange process and ensure more consistent discussion and evaluation conditions. An alternative approach as suggested by P8 could furthermore help to simplify the process: “Define and number the positions to paste the [device] and when it reaches a certain design just tell the users which position to put the plaster on.”

Table 1. Creativity support index (CSI) results for the individual scales.

Scale	Avg. factor Counts (SD)	Avg. factor Score (SD)	Avg. weighted Factor score (SD)
Results worth effort	2.10 (1.70)	15.60 (2.33)	32.5 (27.26)
Exploration	3.70 (1.49)	16.30 (1.85)	61.60 (26.57)
Collaboration	2.50 (1.20)	15.60 (2.33)	39.40 (20.08)
Immersion	2.00 (1.41)	14.40 (2.84)	31.90 (24.16)
Expressiveness	2.50 (1.50)	15.80 (2.56)	38.80 (24.51)
Enjoyment	2.20 (1.33)	16.70 (1.73)	36.40 (22.97)

Exploration scored highest followed by *collaboration* and *expressiveness*. The avg. Factor counts column shows how often participants prioritized a scale within the paired-factor comparisons. The avg. Factor score indicates how high the different scales were rated independent from the priorities (range 0 – 20). The overall CSI score depicts 80.2 (range 0 – 100) and is calculated based on the avg. Weighted factor scores.

7.1.3. Version control for haptic feedback design

When iterating on a design, the new version may not necessary be better than an old version. We also encountered an instance where a design was overwritten and had to be redone. A version control system could prevent this and furthermore allow reverting back to a good state of a design. Also other collaborative features of a version control system like branching and merging could enable new ways of collaboratively working on haptic feedback designs. Commit messages could provide valuable insights in understanding what aspects of a designs result in achieving a goal, and merging designs of multiple designers that individually try

to achieve a specific haptic experience could provide a more direct approach to yield effective haptic feedback designs.

7.2. Collaborative design process

7.2.1. Discussing & evaluating designs

Sharing and discussing the designs provided valuable insights about the designer's ideas of haptic feedback designs. However, we noticed that the flow of presenting and discussing the designs could be improved. As discussed before, explaining the application in detail was sometimes hard without any supporting additional media content helping to understand the application context. Furthermore, participants had to manually load, share, explain and play the design. This was also perceived to be a long process by the participants: "The sharing procedure might be too long, [i.e., the process of] load, share with others, and last play. [...]" (P6) To make the discussions more fluent we believe parts of this process could be better automated. In our case, it was important for participants to discuss the design first before evaluating it as this provided the designer the chance to properly explain the application. However, this also allowed the designer and other collaborators to influence their opinion about the design. While this is not unusual practice in group settings, more objective rating could be achieved when the collaborative toolkit allows collaborators to establish the application context more easily, the designs are presented anonymous in random order and the rating happens before opening the discussion.

7.2.2. Time, iteration count & group sizes

Several choices for the collaborative design process were informed by common group and collaboration techniques and rough estimates since the process of collaborative haptic design significantly differs from e.g., classic pen and paper. As previously mentioned, the focus of the discussions was easily distracted by various factors. Presenting and discussing the designs of five collaborators took longer than the originally anticipated. Instead of the originally planned time of 10 minutes, the discussion phases took around 15 minutes. Long discussion times may lead to participants forgetting about verbal feedback and losing focus and discussion phase times play an important role in forcing quick thinking of creative solutions, but also providing the time necessary to build a distinct idea and exploring the design space. We recommend investigating shorter discussion times than 15 minutes, potentially through a more efficient process and smaller numbers of collaborators to increase the engagement and quality of the designs. Finally, our main focus was to validate if our toolkit could provide the necessary means to enable collaborative effects; hence, two collaboration rounds were sufficient. When refining initial ideas to high-fidelity designs is the goal, an important question to be answered is how many group iteration cycles are enough, or what criteria and threshold should be used to define a stop point for collaborative haptic design.

7.2.3. Rating criteria

Scoring haptic design ideas allows to quickly assess the perceived quality of an idea. In our approach we chose to adapt Goodness of Fit and Goodness of Feel as rating criteria which appeared appropriate as a general evaluation criteria. It provided participants the opportunity to rate the designs once regarding its general feeling, and once regarding its fit to the intended application. We note, that other rating criteria may suit depending on the goal of the session. e.g., if assessing the emotional response of the designs are most interesting, the collaborators could rate the design by the sentiment or regarding specific emotional aspects. Furthermore, we did not incorporate the rating results directly into the design process. Integrating design ratings into the collaboration process could support the collaborating designers to make decisions and help develop more effective designs by e.g., choosing to combine aspects of designs with high ratings. A more structured evaluation approach as proposed by Xu et al. (2014) for visual advertisement design, appear useful to provide more constructive feedback specific to different elements of the design.

7.3. Methodology

While we hope that this work highlights the potential and inspires more work towards collaborative haptic toolkits, we note several methodological limitations that reduces the ability to generalize our results. Part of this is due to the scope of this work, part due the chosen improvable methodological design. In the following, we discuss several of these limitations, which highlight further gaps for future work.

7.3.1. Comparability & quality of solutions

First of all, CoTACS depicts one new instance in the class of collaborative haptic toolkits that focuses on supporting a semi-structured collaborative design process of haptic feedback. A direct comparison with prior toolkits is therefore difficult. However, by evaluating our toolkit via popular evaluation metrics such as the SUS and CSI surveys, we hope to provide a reference for future toolkits to compare with. While we could show that CoTACS could enable interactions and feedback mechanisms that influenced the next iteration of designs, our study was limited in quantifying the quality of the resulting solutions. We recognize that an external evaluation of the collaborative design results and a control group of individual designers could allow to better assess the benefit of collaborative haptic toolkits on the quality of the designs in the future.

7.3.2. Remote collaboration vs. in-person collaboration

In our study, we evaluated our toolkit with a team of remote haptic designers online. We believe that a comparison of remote and in person collaborative design sessions would be valuable to infer if and how collaborative haptic design could be better supported through collaborative toolkits in remote settings.

7.3.3. Semi-structured collaborative design process

For our evaluation, we chose a semi-structured collaboration process that is inspired by common group techniques. However, it only depicts one of many possible ways to explore haptic designs in groups. To the best of our knowledge, ours is the first approach that investigates a semi-structured collaboration process with a collaborative haptic toolkit. While we believe that it depicts an interesting first approach to investigate the potential of the interactions and discussions enabled through our toolkit, we recommend exploring other common techniques that have been used in other domains in the future, such as C-Sketch (Shah et al., 2001). Finally, we note that some prior work indicates that letting individual participants prototype multiple designs before receiving feedback could improve the produced results Dow et al. (2011).

7.4. Application areas

CoTACS was primarily developed to address challenges of designing effective haptic feedback for XR environments such as AR, VR, and MR. Nevertheless, its potential utility isn't limited to these areas and it could be adapted for various other sectors in the future. It could be utilized for creating more effective immersive experiences in gaming and entertainment (Clark et al., 2017; Nakao et al., 2019; Ziat et al., 2014), and for improving the user experience and performance through tactile cues in interfaces and control systems (Ma et al., 2015). In addition, it could be particularly useful to design affective feedback which is usually more susceptible to evoke different experiences in individuals (Papadopoulou et al., 2019; Schneider et al., 2016; Seifi & MacLean, 2013; Umair et al., 2019). Affective feedback designed with CoTACS could provide an additional layer to immersive media content or be used as therapeutic means as explored by prior work (Duvall et al., 2016; Huisman et al., 2013; Schelle et al., 2015; Stanley & Kuchenbecker, 2011; Yarosh et al., 2017). It also holds potential for designing more effective haptic feedback for assistive technology, e.g., supporting rehabilitation (Bark et al., 2009; Keltys-Stephen & Dixon, 2013; Likens et al., 2020). Finally the potential benefit of CoTACS for applications in telemedicine, skill transfer in sports, aviation training, or as an educational tool could be explored in the future.

8. Conclusion

In this work, we proposed a new toolkit, CoTACS, to address challenges of haptic feedback design by enabling haptic designers to collaborate and explore effective haptic feedback together through semi-structured group collaboration. Our toolkit extends prior work by addressing limitations such as the "inability to tweak," allowing users to easily reuse and iterate on each others designs. In our semi-structured haptic collaboration process, we introduce a first group evaluation process. Our findings showed that our collaborative haptic toolkit could be used by users with different levels of expertise to explore on-body haptic feedback for new emerging

XR environments. Our toolkit enabled synergistic effects through the interactions between the designers during the group collaboration task and allowed inspiring new ideas. We saw that it successfully enabled designers to reuse and iterate on each other's designs and we demonstrate how the rating of designs allows to quickly identify better designs based on group feedback. We discussed several limitations of our toolkit and elaborated on multiple starting points for future collaborative haptic toolkits to support exploring haptic feedback in groups. We hope that our approach inspires more work towards collaborative haptic toolkits and design practices that help to explore haptic feedback designs more effectively.

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Ethical approval

The studies of this work have been carried out in accordance with approved guidelines and regulations. Ethics approval for the research was obtained from the University of Auckland Human Participants Ethics Committee, The University of Auckland (Ref. UAHPEC1470).

Disclosure statement

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